Software Quality and Project Development

Università degli Studi di Catania
Dipartimento di Matematica e Informatica



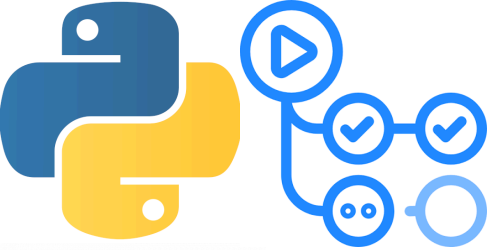
$ whoami

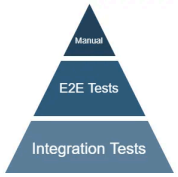
Hi, I am Stefano Borzì 👋

Open-source developer

Full Stack Developer at "Royal BAM Group"

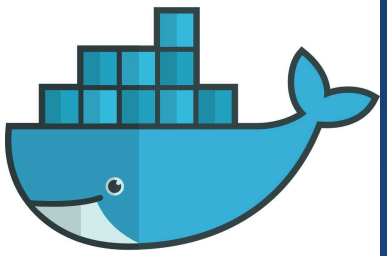
https://stefanoborzi.dev/



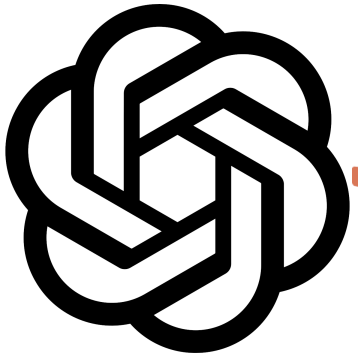






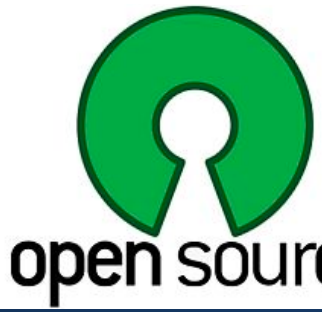














Intro

- Shell UNIX
- Git
- GitHub
- Opensource
- Python
- Unit-test
- SOLID principles
- E2E tests with playwright
- CI / CD pipeline & tools
- Docker
- AI & Agentic coding



1991

Python

S.O.L.I.D.

2000–2004

SOLID Principles



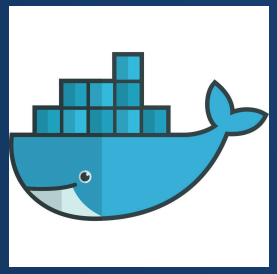
2005

Git

GitHub

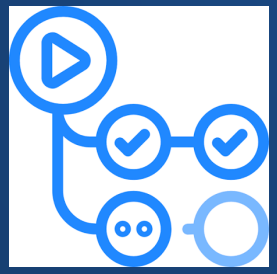
2008

GitHub



2013

Docker



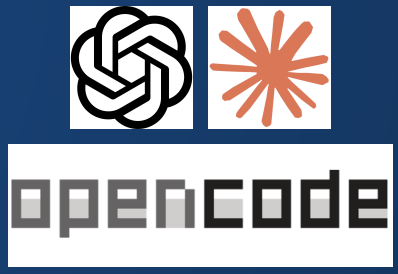
2018

GitHub Actions



2020

Playwright



2023–2025

Agentic Coding

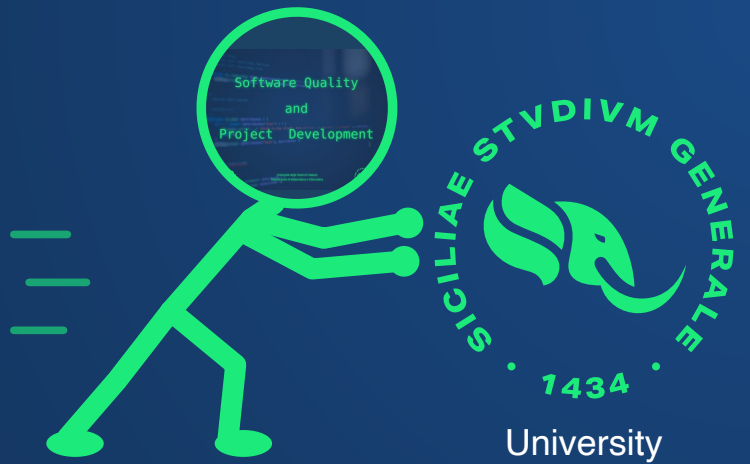


University

knowledge gap



Work world



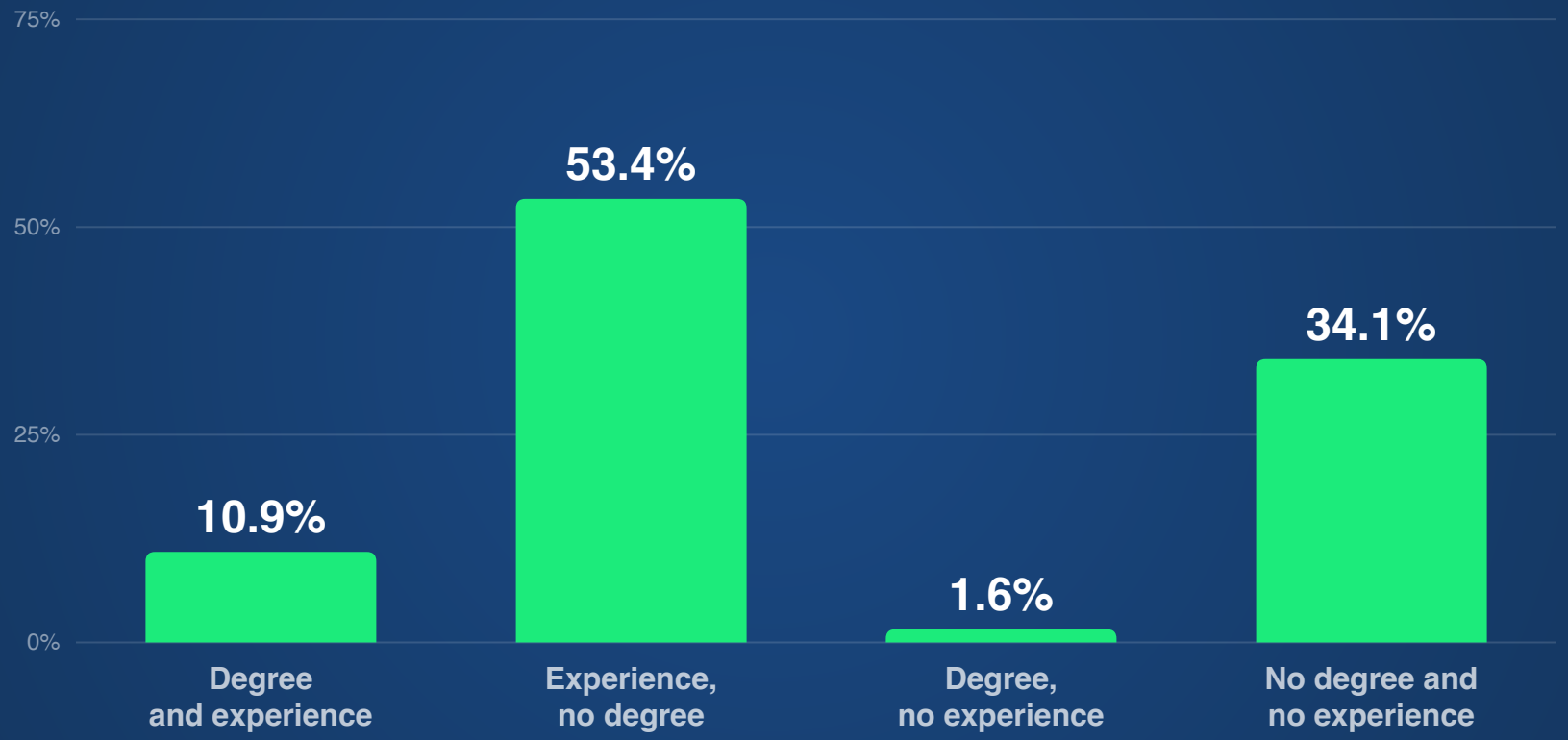
University



Work world

Experience, degree, or both? What Italian companies ask for

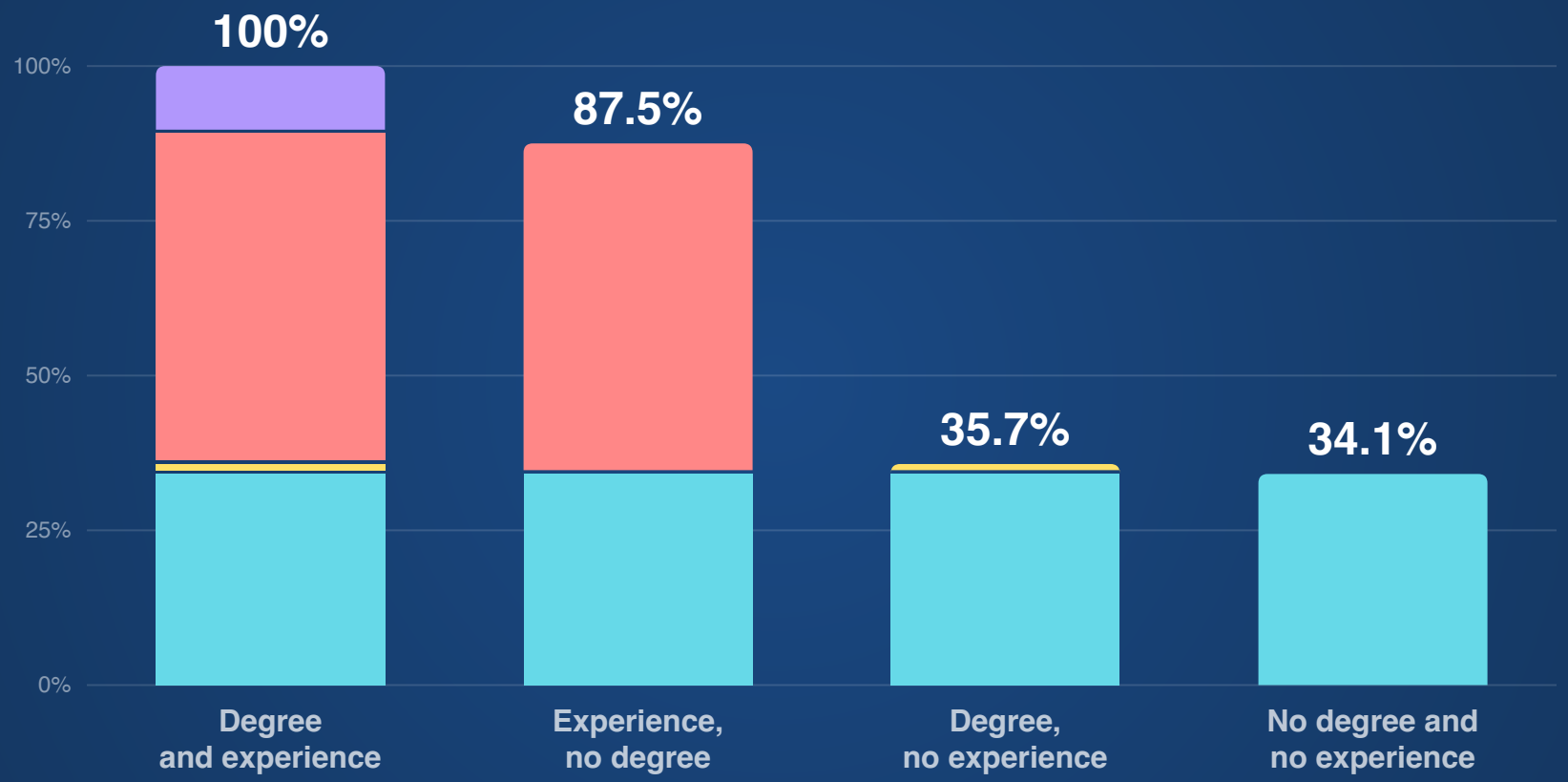
Share of all planned hires in Italy in 2024 (5.5 million)



Source: Unioncamere, Sistema Informativo Excelsior 2024: https://excelsior.unioncamere.net/sites/default/files/pubblicazioni/2024/Domanda_professioni_formazione_imprese.pdf p. 120 Table 16: 5,516,280 planned hires, 3,546,440 of them require experience · p. 122 Table 18: 691,090 require a university degree, 87.2% of them also require experience

Experience, degree, or both? What Italian companies ask for

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Role

As a senior software developer, you will be responsible for designing solutions to problems from the group up, reviewing code from members of your team, TDD all of your code, help teammates improve in the technical skills and knowledge, whilst also working closely with product managers and company wide stakeholders. You will be part of an enthusiastic team responsible for product development of platform services, which is mostly high scale backed services. You will have the ability to choose the right tools and methods that you decide are appropriate to solve any problems, but will be supported by experienced team members.

Senior Software Developer Requirements:

- 3 years experience in Javascript and/or Typescript
- Experience with React framework
- Proficiency with Git source control
- Efficient HTML and CSS ability
- Good communication skills in written, verbal, and code

Offer:

- Up to £90,000

WHAT WE'D LOVE YOU TO DO (AND LOVE DOING)

- Develop high-quality and responsive user interfaces for web applications using modern frontend frameworks such as React, Angular.
- Collaborate with designers to create visually appealing interfaces that meet user needs and brand guidelines
- Work closely with product managers and backend engineers to integrate frontend components with backend APIs and databases
- Ensure the web applications are optimized for speed and performance, including minimizing load times and maximizing interactivity
- Write clean, modular, and maintainable code that can be easily understood and modified by other team members
- Participate in code reviews and provide constructive feedback to other team members
- Stay up-to-date with the latest frontend technologies and trends and bring new ideas and solutions to the team
- Ensure the web applications are accessible to all users, including those with disabilities
- Test and debug frontend components to identify and fix issues that arise during development
- Document technical specifications and provide support to other team members as needed

WE'D LOVE TO MEET YOU IF

Must Haves-

- Strong proficiency in JavaScript, TypeScript, HTML, CSS, and Node.js
- Experience with modern frontend frameworks such as React
- Strong problem-solving skills and attention to detail
- Excellent communication and collaboration skills
- Experience with version control tools such as Git
- Commitment to continuous learning and self-improvement

🔥 **What Challenges A**

- Manage, lead and c
- Design, develop, ar
- Create high-quality
- Help our analysts w face
- Develop new featur systems
- Take ownership of p

👉 **Who You Are**

- You have a degree in Compute equivalent, or relevant experie
- You have outstanding Python
- ...with at least 5 years of profe
- You have experience with Doc
- You have a good understandin
- You know when to use NoSQL RabbitMQ
- You feel comfortable building
- You know your way around so documentation, and continuou
- ...and you don't mind collectin Kubernetes skills

- minimo di 4 anni esperienza con PHP e MySQL;
- minimo 3 anni di esperienza con Laravel;
- minimo 2 anni di esperienza con Vue.js;

Senior Software Engineer, Data Infrastruct...

Vinted · Amsterdam, North Holland, Netherlands (H...

Apply Save ...

In this position, you'll

- Work within a team and participate in decision-making, planning and implementing data infrastructure solutions that improve our users' experience
- Develop scalable, distributed, cost-effective solutions for data loading, processing, transformation and analysis
- Collaborate with other developers through GitHub pull requests and code reviews
- Write and maintain IaC (Terraform, Terragrunt)
- Implement and adopt DevOps practices in our platform
- Monitor and troubleshoot platform issues. We monitor our code and infrastructure through Kibana, Grafana and other tools
- Consult DSA and other engineering teams on best practices and implementation
- Apply Agile and Lean principles in your work
- Write unit tests for the code you write

Our Tech Stack

- **Backend Code:** Rust, Tokio, Cargo, Git (https://github.com/cowprotocol/services)
- **Infrastructure:** Pulumi, AWS, Docker, Kubernetes, Grafana, Prometheus, Elasticsearch
- **Other codebases:** Typescript, Python, Solidity

Who you are

- You have 3+ years of experience with system programming, preferably in Rust (C++ or similar)
- You have excellent computer science, programming, and algorithmic skills
- You care deeply about the quality, readability, observability, and testability of your code
- You proactively give technical direction to improve performance, prevent issues, optimise data us
- You are a team player, a strong communicator, and gladly share your expertise with others
- You collaborate effectively with a remote-first team on a large, open-source codebase
- Excited to explore and gain experience with Ethereum

Your skills

To begin with, we are looking for junior Data Scientists with the following essential skills and experiences:

- Master of Science degree in Mathematics, Statistics, Computer Science, Engineering Physics, Electrical Engineering or related fields.
- Programming skills with one or more of the following languages: Python, R, Scala.
- Theoretical background in Machine Learning, Statistical Data Analysis, and/or Optimization.
- Great communication skills and good at explaining complex technical concepts to business leaders in a compelling way.

It is meritorious if you also have some of the following experiences or skills:

- Hands-on experience of implementing analytics & machine learning projects.
- Hands-on experience of querying data from databases and big data platforms.
- Data visualization with Tableau, Power BI or other business intelligence tools.
- DevOps for data & analytics: Git, MLflow, Docker, Kubernetes, and CI/CD tools (Jenkins, GitHub Actions)
- Experience of working with Cloud platforms, for example MS Azure, GCP, AWS, and Databricks.

I graduated in Computer Science in the early 2000s.

When I took a Databases class, NoSQL didn't exist.
When I took a Computer Graphics class, OpenGL didn't support shaders.
When I took a Computer Security class, no one knew about botnets yet.
When I took an Artificial Intelligence class, deep learning didn't exist.
When I took a Programming Languages class, reactive programming wasn't a "thing".
When I took a Distributed Systems class, there was no Big Data or cloud computing.
When I took an Operating Systems class, hypervisors didn't exist (in PCs at least).
When I took a Networking class, there was no wifi in my laptop or internet in my phone.

Learn the fundamentals. The rest will change anyway.

Exam

A project to be done in a group of two people **(no more).**



